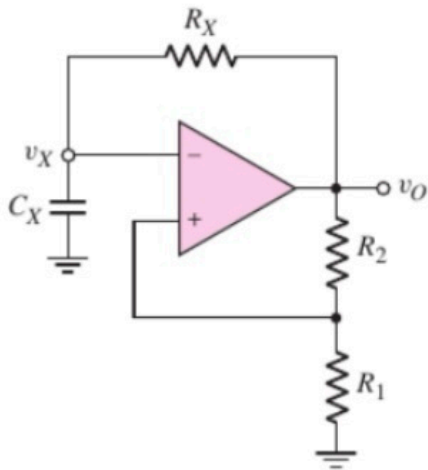


Square Wave Generator

1.



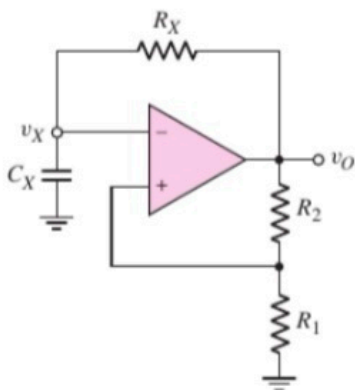
Here suppose $R_1 = 10\text{k}$, $R_2 = 20\text{k}$, $R_X = 1\text{k}$, $C_X = 1\text{ mF}$, $V_H = 10\text{V}$ and $V_L = -10\text{ V}$

a. Find the period and frequency of the square wave?

b. What will be the value of the duty cycle of the square wave?

c. Draw the output waveform with proper labeling.

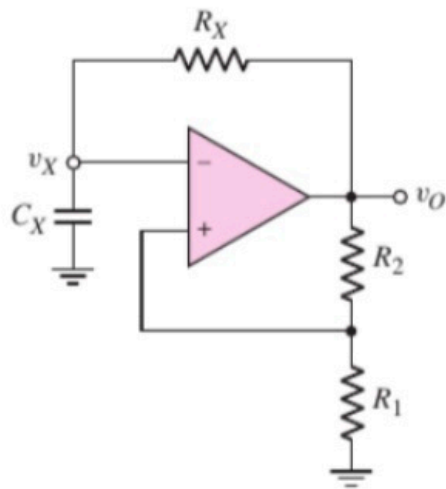
2.



Here $V_H = 10\text{ V}$ and $V_L = -10\text{ V}$

Design the circuit so that it can generate a square wave with 1 kHz frequency and 50% duty cycle.

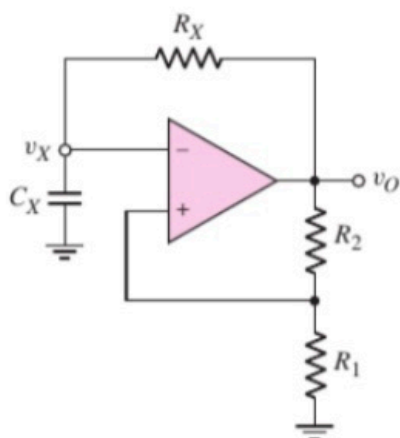
3.



suppose $R_1 = 10k$, $R_2 = 20k$, $R_X = 1k$, $C_X = 1 \text{ mF}$

Design the circuit so that it can generate square wave with a 30% duty cycle. Find the frequency of your designed circuit.

4.



Here suppose $R_1 = 10k$, $R_2 = 20k$, $R_X = 1k$, $C_X = 1 \text{ mF}$, $V_H = 10V$ and $V_L = -5V$

Find out the duty cycle of inverted output signal of the above circuit.

1. a) $R_1 = 10k, R_2 = 20k, R_x = 1k, C_x = 1mF,$

$$V_H = +10V, V_L = -10V$$

$$V_{TH} = \cancel{10} V_H \times \frac{R_1}{R_1 + R_2}$$

$$= 10 \times \frac{10}{10 + 20} = 3.33$$

$$V_{TL} = V_L \times \frac{R_1}{R_1 + R_2} = -3.33$$

$$\tau = R_x C_x = 1k \times 1m = 1s$$

$$\begin{aligned} T_1 &= \tau \ln \frac{V_H - V_{TL}}{V_H - V_{TH}} \\ &= 1 \times \ln \frac{10 - (-3.33)}{10 - 3.33} \\ &= 0.69s \end{aligned}$$

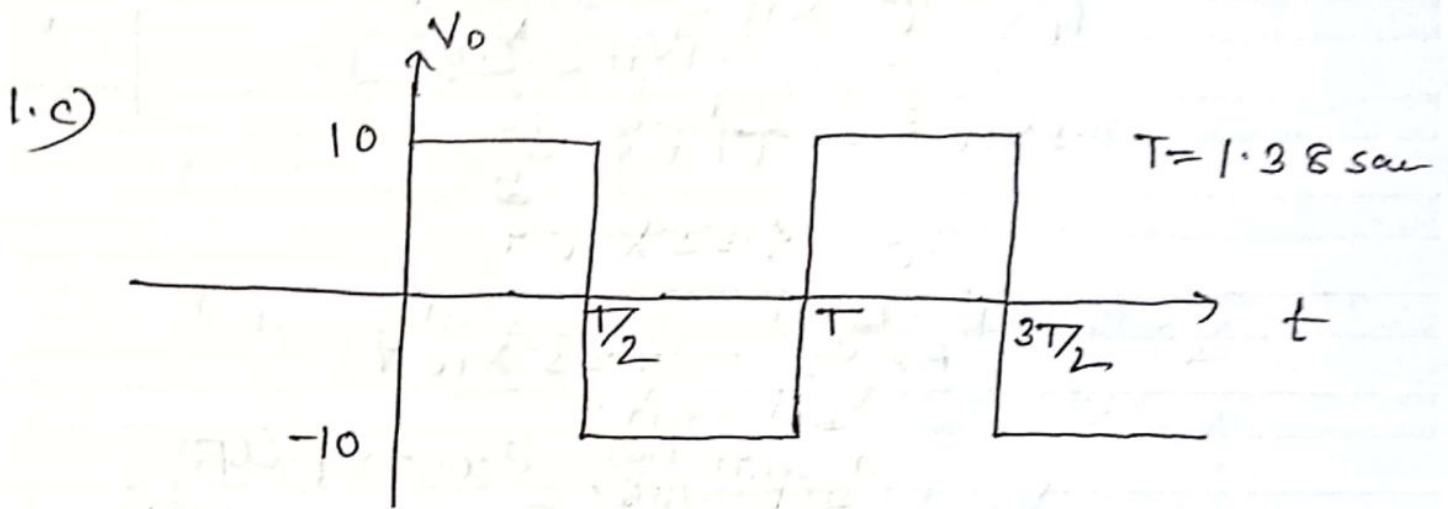
$$\begin{aligned} T_2 &= \tau \ln \frac{V_L - V_{TH}}{V_L - V_{TL}} \\ &= 1 \times \ln \frac{-10 - 3.33}{-10 - (-3.33)} \\ &= 0.69s \end{aligned}$$

$$T = T_1 + T_2 = 0.69 + 0.69 = 1.38s$$

$$f = \frac{1}{T} = 0.72 \text{ Hz}$$

$$1. b) \text{ Duty cycle} = \frac{T_1}{T_1 + T_2} \times 100\%$$

$$= \frac{0.69}{0.69 + 0.69} \times 100\% \\ = 50\%$$



$$2. \quad V_H = 10\text{V}, \quad V_L = -10\text{V}$$

Assuming, $R_1 = R_2 = 10\text{k}\Omega$

$$V_{TH} = V_H \times \frac{R_1}{R_1 + R_2} = 5\text{V}$$

$$V_{TL} = V_L \times \frac{R_1}{R_1 + R_2} = -5\text{V}$$

$$\text{given, } f = 1\text{kHz}, \quad T = \frac{1}{f} = 1\text{ms}$$



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Duty cycle = 50% and, $T = T_1 + T_2$

$$\frac{T_1}{T_1 + T_2} = 0.5, \Rightarrow T_1 = T_2 = \frac{T}{2} = 0.5 \times 10^{-3}$$

$$T_1 = \tau \ln \frac{V_H - V_{TL}}{V_H - V_{TH}}$$

$$0.5 \times 10^{-3} = \tau \ln \frac{15}{5}$$

$$\tau = 4.55 \times 10^{-4}$$

$$R_x C_x = 4.55 \times 10^{-4}$$

assuming, $C_x = 1 \mu\text{F}$

$$R_x = \frac{4.55 \times 10^{-4}}{C_x} = 455 \Omega$$



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$$2. \quad \tau = R \times C = 1 \mu \times 1 \text{mF} = 1 \text{s}$$

$$\frac{\tau_1}{\tau_1 + \tau_2} = 0.3 = \frac{3}{10}$$

$$\frac{\tau_1}{\tau_2} = \frac{3}{7}$$

Assuming $R_1 = R_2 = 10$

$$\tau_1 = \tau \ln \frac{V_H - V_L}{V_H - V_{TH}}$$

$$\tau_1 = \tau \ln \frac{V_H - V_L \frac{R_1}{R_1 + R_2}}{V_H - V_H \frac{R_1}{R_1 + R_2}}$$

$$\tau_1 = \tau \ln \frac{V_H - V_L/2}{V_H - V_H/2}$$

$$\tau_1 = \tau \ln \frac{2V_H - V_L}{V_H}$$

$$\tau_2 = \tau \ln \frac{V_L - V_{TH}}{V_L - V_{TL}}$$

$$= \tau \ln \frac{V_L - V_H/2}{V_L}$$

$$\tau_2 = \tau \ln \frac{2V_L - V_H}{V_L}$$



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Now, $7T_1 = 3T_2$

$$\Rightarrow 7 \ln \frac{2V_H - V_L}{V_H} = 3 \ln \frac{2V_L - V_H}{V_L}$$

suppose, $V_H = 10V$,

Find $V_L = ? - 2.24$ (use calculator)
solve eqn

4. $\tau = R \times C_x = 1s$

$V_H = +10V$, $V_L = -5V$

$$V_{TH} = V_H \frac{R_1}{R_1 + R_2} = 10 \times \frac{10}{10 + 20} = 3.33$$

$$V_{TL} = V_L \frac{R_1}{R_1 + R_2} = -5 \times \frac{10}{10 + 20} = -1.67$$

$$T_1 = \tau \ln \frac{V_H - V_{TL}}{V_H - V_{TH}} = 1 \times \ln \frac{10 - (-1.67)}{10 - 3.33} = 0.559$$

$$T_2 = \tau \ln \frac{V_L - V_{TH}}{V_L - V_{TL}} = 1 \times \ln \frac{-5 - 3.33}{-5 - (-1.67)} = 0.913s$$

Duty cycle = $\frac{T_1}{T_1 + T_2} \times 100\% = 37.9\%$